

# JOEL PEPPER

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Philadelphia, PA

## EDUCATION

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### **Drexel University**

*September 2019 – June 2026*

Ph.D. in Computer Science [2]  
M.S. in Computer Science  
Advisor: [Dr. David E. Breen](#)  
Member of Upsilon Pi Epsilon  
Overall GPA: 3.99

### **The Ohio State University**

*August 2014 – August 2019*

B.S. in Biomedical Engineering  
B.S. in Computer Science & Engineering  
Member of Tau Beta Pi  
Cum Laude (BME Degree)  
Honors in Engineering (CSE Degree)  
Overall GPA: 3.485

## RESEARCH STATEMENT

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**Fields of interest:** Computational Biology, Bioinformatics, Genomics, Biomedical Image Analysis

My thesis research was on the modeling and simulation of biological and bio-inspired structures. I developed a simulation framework for “4D materials,” which are materials that exhibit an active, time dependent warping response to some stimuli. Examples of 4D materials include epithelial tissue, 4D prints (3D prints that warp when heated), virtual creatures and soft robots. My first publication on this topic detailed a novel 4D printing configuration involving multi-layered, cuboidal grids [11]. My second dissertation publication involved an extension of this framework to an epithelium-inspired grid of hexagonal prisms [5]. A chapter in my thesis also covers an extension of the epithelium modeling capabilities to include cell-level, neural network-based agency, in order to enable cells to actively respond to their surroundings in a more biomimetic fashion [2].

Other areas of research I have been involved in outside of my dissertation project include cell and tissue engineering, image analysis, and genomics.

## FUNDING

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### **NSF Graduate Research Fellowship**

*September 2021 – September 2024*

Proposal was entitled “Generalized Agent-Based Model of Responsive 4D Materials” [2], [5], [11]

## EMPLOYMENT HISTORY

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### **Drexel Univ Dept of Computer Science**

*January 2026 – March 2026*

*Doctoral Teaching Fellow*

*Philadelphia, PA*

- Assisted in instruction and graded for an undergraduate operating systems course

### **Drexel Univ Metadata Research Center**

*September 2024 – December 2025*

*Doctoral Research Fellow*

*Philadelphia, PA*

- Funded researcher on several projects under the direction of [Dr. Jane Greenberg](#)
- Institute for Data-Driven Dynamical Design: NSF-sponsored “Harnessing the Data Revolution” (HDR) institute working to automatically extract and study the contents of paper lab notebooks [4], [6]

- Major Research Instrumentation: Accessible Data-Intensive Science: NSF-sponsored project to develop a platform for accessible data-intensive science and engineering; responsibilities included working to improve platform's data processing pipeline to seamlessly move data from scientific instruments to a distributed data management and analysis system as they are generated
- HIVE-4-MAT: Undertook a full rewrite and modernization of the "Helping Interdisciplinary Vocabulary Engineering" (HIVE) web application (release forthcoming)

### **Drexel Univ/Self**

*Doctoral Fellow*

*September 2021 – August 2024*

*Philadelphia, PA*

- Externally funded via National Science Foundation Graduate Research Fellowship [2], [5], [11]

### **Memorial Sloan Kettering Cancer Center**

*Computational Biology Intern*

*June 2023 – September 2023*

*New York, NY*

- Research intern studying spatial epigenomics data under Dr. Christina Leslie
- Focus of research was on analyzing 3D chromosomal imaging data to look for novel interaction patterns which may correspond with genetic disorders

### **Drexel Univ Graduate College**

*Remote Course Facilitator*

*September 2020 – December 2021*

*Philadelphia, PA*

- Assisted professors with the technical and logistical aspects of running online courses for four quarters during the pandemic

### **Biology Guided Neural Networks Project**

*Doctoral Research Fellow*

*September 2020 – September 2021*

*Philadelphia, PA*

- Funded researcher (under Dr. Jane Greenberg) on an NSF HDR project working on automatic generation of missing/deficient metadata for digital specimen images [3], [7], [8], [9], [10], [12]

### **Drexel Univ Digital Development Camp**

*Counselor*

*August 2020 – August 2020*

*Philadelphia, PA*

- Lead counselor for a web development camp sponsored by the college's Women in Tech Initiative

### **Drexel Univ Dept of Computer Science**

*Doctoral Teaching Fellow*

*September 2019 – August 2020*

*Philadelphia, PA*

- Assisted in instruction and graded for graduate (CS foundations, advanced programming techniques) and undergraduate (freshman design, data structures) courses both in person and online

### **Drexel Univ RETHink**

*Graduate Assistant*

*July 2020 – July 2020*

*Philadelphia, PA*

- Assisted in the computer science department's NSF-sponsored Research Experience for Teachers

### **Lab-Ally**

*Contract Software Developer*

*November 2017 – September 2019*

*Columbus, OH*

- Developed software for an electronic lab notebook (ELN), focusing mainly on the server component
- Worked to modernize and improve the reliability and readability of a large legacy codebase
- Made significant improvements to the development operations pipeline for the ELN server

### **OSU Dept of Computer Science and Engineering**

*Student Instructional Assistant*

*August 2016 – July 2019*

*Columbus, OH*

- Graded for and assisted in instruction of a web development precapstone course

- Worked on projects in nanotechnology and cell migration [13]
- Experimental protocols performed included cell migration studies; cell fixation, staining and imaging; traditional and 3D hydrogel cell culturing; and electrospray ionization

## AWARDS

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- (2023) College of Computing and Informatics Outstanding Graduate Student Award
- (2023) Phoebe Haas Doctoral Endowed Scholarship from the College of Computing and Informatics
- (2021) Best Student Paper Award for “Automatic Metadata Generation for Fish Specimen Image Collection” [12]
- (2020) Highly Commended/Honorable Mention for Drexel College of Computing and Informatics Graduate Teaching Assistant Excellence Award
- (2018) OSU College of Engineering Excellence in Capstone Design Award [14]

## PUBLICATIONS

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- [1] S. McClellan, A. Ireland, C. Gerber, J. Pepper, C. B. Rauch, et al., “MatSci-YAMZ: Integrating Human and Artificial Intelligence Vocabulary,” en, *Data Science Journal*, volume 25, number 1, Jul. 2026, ISSN: 1683-1470. DOI: [10.5334/dsj-2026-023](https://doi.org/10.5334/dsj-2026-023)
- [2] J. Pepper, “An epithelium-inspired deformation modeling framework for responsive 4d materials,” Ph.D. dissertation, Drexel University, Jun. 2026, xxii, 149 pages. DOI: [10.17918/00011498](https://doi.org/10.17918/00011498)
- [3] J. Greenberg, J. Pepper, X. Zhao, R. Marciano, D. Breen, and Y. An, “The metadata ecosystem and AI: Enabling FAIR and AI-ready data,” en, *AI Magazine*, volume 47, number 2, e70060, Jun. 2026, ISSN: 2371-9621. DOI: [10.1002/aaai.70060](https://doi.org/10.1002/aaai.70060)
- [4] J. Pepper, Z. Siapno, J. Furst, F. Uribe-Romo, D. Breen, and J. Greenberg, “From Analog Records to Computational Research Data: Building the AI-Ready Lab Notebook,” in *2025 IEEE International Conference on Big Data (BigData)*, Dec. 2025, pages 6018–6023. DOI: [10.1109/BigData66926.2025.11401845](https://doi.org/10.1109/BigData66926.2025.11401845)
- [5] J. Pepper and D. E. Breen, “An Epithelium-Inspired Deformation Modeling Framework for 4D Sheets,” in *Advances in Visual Computing*, Springer, Jan. 2025, pages 111–124, ISBN: 978-3-031-77392-1. DOI: [10.1007/978-3-031-77392-1\\_9](https://doi.org/10.1007/978-3-031-77392-1_9)
- [6] J. Pepper, E. Jones, X. Zhao, J. Furst, K. Langlois, et al., “AI-Ready Data: Knowledge Extraction from Archival Lab Notebooks,” in *2024 IEEE International Conference on Big Data (BigData)*, Dec. 2024, pages 2489–2495. DOI: [10.1109/BigData62323.2024.10825206](https://doi.org/10.1109/BigData62323.2024.10825206)
- [7] M. A. Balk, J. Bradley, M. Maruf, ..., J. Pepper, et al., “A FAIR and modular image-based workflow for knowledge discovery in the emerging field of imageomics,” *Methods in Ecology and Evolution*, Apr. 2024, ISSN: 2041-210X. DOI: [10.1111/2041-210X.14327](https://doi.org/10.1111/2041-210X.14327)
- [8] D. E. Breen, A. Senin, A. Levere, J. Pepper, and J. Greenberg, “Specimen Outlining: A Computational Archival Science Approach,” in *2023 IEEE International Conference on Big Data (BigData)*, Dec. 2023, pages 2004–2009. DOI: [10.1109/BigData59044.2023.10386985](https://doi.org/10.1109/BigData59044.2023.10386985)
- [9] J. Pepper, A. Senin, D. Jebbia, D. Breen, and J. Greenberg, “Metadata Verification: A Workflow for Computational Archival Science,” in *2022 IEEE International Conference on Big Data (Big Data)*, Dec. 2022, pages 2565–2571. DOI: [10.1109/BigData55660.2022.10020340](https://doi.org/10.1109/BigData55660.2022.10020340)
- [10] K. Karnani, J. Pepper, Y. Bakış, X. Wang, H. Bart Jr., D. E. Breen, and J. Greenberg, “Computational metadata generation methods for biological specimen image collections,” en, *International Journal on Digital Libraries*, Nov. 2022, ISSN: 1432-1300. DOI: [10.1007/s00799-022-00342-1](https://doi.org/10.1007/s00799-022-00342-1)

- [11] J. Pepper and D. Breen, “Design and Simulation of Multi-Tiered 4D Printed Objects,” *Computer-Aided Design and Applications*, volume 20, number 3, pages 489–506, Sep. 2022. DOI: [10.14733/cadaps.2023.489-506](https://doi.org/10.14733/cadaps.2023.489-506)
- [12] J. Pepper, J. Greenberg, Y. Bakiş, X. Wang, H. Bart, and D. Breen, “Automatic Metadata Generation for Fish Specimen Image Collections,” in *2021 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, Sep. 2021, pages 31–40. DOI: [10.1109/JCDL52503.2021.00015](https://doi.org/10.1109/JCDL52503.2021.00015)
- [13] Y. Cui, S. Cole, J. Pepper, J. J. Otero, and J. O. Winter, “Hyaluronic acid induces ROCK-dependent amoeboid migration in glioblastoma cells,” en, *Biomaterials Science*, volume 8, number 17, pages 4821–4831, Aug. 2020, ISSN: 2047-4849. DOI: [10.1039/D0BM00505C](https://doi.org/10.1039/D0BM00505C)

## In Review

- Middle authorship, “Yarn-Level Simulation of the Mesoscale Self-Folding Behavior of Weft-Knitted Textiles,” World Conference of the Association of Universities for Textiles Proceedings 2026

## PATENT

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- [14] Z. Ali, P. Scott, A. Kiourti, B. Zhang, J. Pepper, et al., “System and method for completing a measurement of anxiety,” Granted Patent US 11571155 B2, Feb. 7, 2023. [Online]. Available: <https://lens.org/025-853-906-337-42X>

## SOURCE CODE

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- Source code for [2], [5], [11]: [DrJPepper/4d-multilayer-modeling](https://github.com/DrJPepper/4d-multilayer-modeling)
- Source code for [9], [10], [12]: [hdr-bgnn/drexel\\_metadata](https://github.com/hdr-bgnn/drexel_metadata)

## PRESENTATIONS

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- Workshop paper presentation, “From Analog Records to Computational Research Data: Building the AI-Ready Lab Notebook,” Computational Archival Science Workshop @ IEEE Big Data, Online, December 2025 [4] ([Original Video Link](#), [Mirror](#), [Slides](#))
- Workshop presentation, “Automatic Metadata Generation and Verification for AI-Ready Fish Image Collections,” Imageomics Workshop @ The Association for the Advancement of AI Conference, Philadelphia, PA, March 2025 ([Poster](#))
- Workshop paper presentation, “AI-Ready Data: Knowledge Extraction from Archival Lab Notebooks”, Computational Archival Science Workshop @ IEEE Big Data, Washington, DC, December 2024 [6] ([Original Video Link](#), [Mirror](#), [Slides](#))
- Paper presentation, “An Epithelium-Inspired Deformation Modeling Framework for 4D Sheets”, International Symposium on Visual Computing, Stateline, NV, October 2024 [5] ([Recorded Practice Run](#))
- Poster presentation given by REU student, “AI-Ready Data: Knowledge Extraction from Laboratory Notebooks,” Northwestern University ID4 Symposium, Evanston, IL, August 2024 ([Poster](#))
- Poster presentation, “An Epithelium-Inspired Deformation Modeling Framework for 4D Sheets,” Drexel Emerging Graduate Scholars Conference, Philadelphia, PA, April 2024 ([Poster](#))
- Workshop presentation, “Enumerating and Addressing AI-Readiness Challenges from the Biology Guided Neural Networks HDR Project,” AI-Ready Data: Navigating the Dynamic Frontier of Metadata and Ontologies, Philadelphia, PA, April 2024 ([Slides](#))
- Hackathon participant, “Evaluating the effects of Hi-C and TF on cis-regulation of gene expression,” 4D Nucleome Hackathon, Seattle, WA, March 2024 ([Presentation Slides](#))

- Workshop paper presentation, “Metadata Verification: A Workflow for Computational Archival Science” (with Andrew Senin), Computational Archival Science Workshop @ IEEE Big Data Conference, Online, December 2022 [9] ([Video](#))
- Paper presentation, “Design and Simulation of Multi-Tiered 4D Printed Objects,” Computer-Aided Design and Applications Conference, Online, July 2022 [11] ([Recorded Practice Run](#))
- Workshop, “3D Data Visualization with VTK,” Philly Codefest (run by Drexel University), Online, March 2022 ([Original Video Link](#), [Mirror](#), [Demo Code](#))
- Paper presentation, “Automatic Metadata Generation for Fish Specimen Image Collections,” ACM / IEEE Joint Conference on Digital Libraries, Online, September 2021 [12] ([Video](#))
- Presenter and panelist, Taxonomic Names and Metadata: A Framework for Big Data Interoperability Session, Great Lakes Bioinformatics Conference, Online, May 2021
- Talk, “Approaches for Computing Specimen Image Research Data” (with Jane Greenberg and David Breen), International Conference on Statistical Tools and Techniques for Research Data Analysis, Online, January 2021

## MENTORING

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- [Zach Siapno](#), Undergraduate Researcher, March 2025 – September 2025 [4]
- [Elizabeth Jones](#), Summer REU, June 2024 – August 2024 ([Poster](#))
- [Andrew Senin](#), Undergraduate Researcher and Senior Design Student, August 2022 – December 2023 [8], [9]
- [Kevin Karnani](#), Undergraduate Research Co-Op, September 2021 – March 2022 [10]

## TECHNICAL STRENGTHS

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|---------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Environment</b>        | GNU/Linux user of 15+ years with extensive knowledge of its system administration, networking and scripting |
| <b>Computer Languages</b> | C++, C, Java, Python, HTML/CSS, JavaScript, SQL, Bash                                                       |
| <b>Tools   Libraries</b>  | Git, Vim, Make, CMake   PyTorch, OpenCL, VTK, Qt, Svelte                                                    |